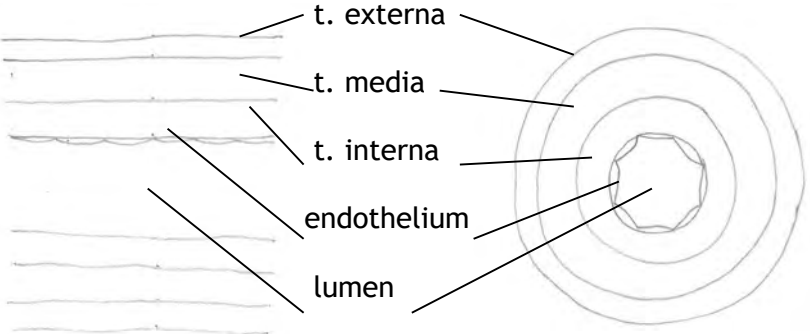


Question number	Answer	Additional guidance	Mark
7(a)	answer to no more than one decimal place in the range of 1.3 to 2.4 (m)		(1)

Question number	Answer	Additional guidance	Mark
7(b)	An explanation that includes two of the following points: • because blood has to be pumped under high pressure (from the heart) (1) • so that the blood can reach the { brain / head} (1) • pump blood against (the force of) gravity (to the upper parts of the body) (1)		(2)

Question number	Answer	Additional guidance	Mark
7(c)(i)	A drawing that shows : {3 layers / 3 layers and endothelium} + lumen (1) Any two from: - lumen (1) - endothelium / epithelium (1) - tunica {interna / intima} (1) - tunica media (1) - tunica {externa / adventitia} (1)	 ACCEPT inside line labelled if 2 layer diagram with no endothelium labelled IGNORE smooth muscle ACCEPT (smooth) muscle (cells) and elastic fibres ACCEPT collagen fibres N.B. If one incorrect label, max of 1 label mark If two incorrect labels then no label marks can be awarded	(3)

Question number	Answer	Additional guidance	Mark
7(c)(ii)	An explanation that includes the following points: - the blood is under high pressure so that the {arteries need to widen / elastic fibres need to stretch} (1) - elastic recoil necessary to maintain the high blood pressure (1)	N.B. need to stretch and recoil to {maintain / accommodate} high pressure = 1 mark if no other mps awarded	(2)

Question number	Answer	Additional guidance	Mark
7(d)(i)	An explanation that includes the following points: - because there will be less blood flowing (near the surface of the skin at any one time) (1) - therefore blood clot will form {more easily / faster} (1)	ACCEPT blood at {lower / low} pressure ACCEPT clotting will be {faster / easier} less blood lost whilst clot forming	(2)

Question number	Answer	Additional guidance	Mark
7(d)(ii)	An explanation that includes the following points: • because (the smaller red blood cells will have) a greater surface area (1) • and therefore oxygen will be able to diffuse (into / out of the RBCs) faster (1)	IGNORE easier to diffuse / more can diffuse	(2)